

# HTWK Traffic Safety Knowledge Graph Saxony

## Linking speed cameras, accidents and regional data

Martin Gutwasser & Sebastian Neubert – HTWK Leipzig – Semantic Web Project 2026

### 1 MOTIVATION

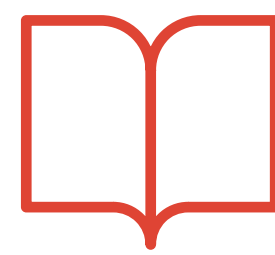
- Traffic safety depends on many complex factors.
- Open datasets are available, but they are often hard to compare directly.
- Semantic Web technologies allow us to connect these datasets in a common format.

### 2 DATA SOURCES



#### OpenStreet Map

fixed speed cameras,  
coordinates, direction,  
max speed via  
**Overpass turbo**



#### Unfallatlas

accident locations  
with personal injury,  
year, coordinates,  
...



#### Open Data Sachsen

municipalities,  
population,  
area, ...

### 3 WORKFLOW



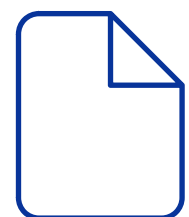
#### Open Data

open data portals  
(CSV, GeoJSON, ...)



#### Python Extraction

clean, transform



#### RDF / Turtle

convert to RDF  
(Turtle syntax)



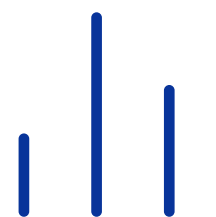
#### Triple Store

store and connect  
RDF data



#### SPARQL

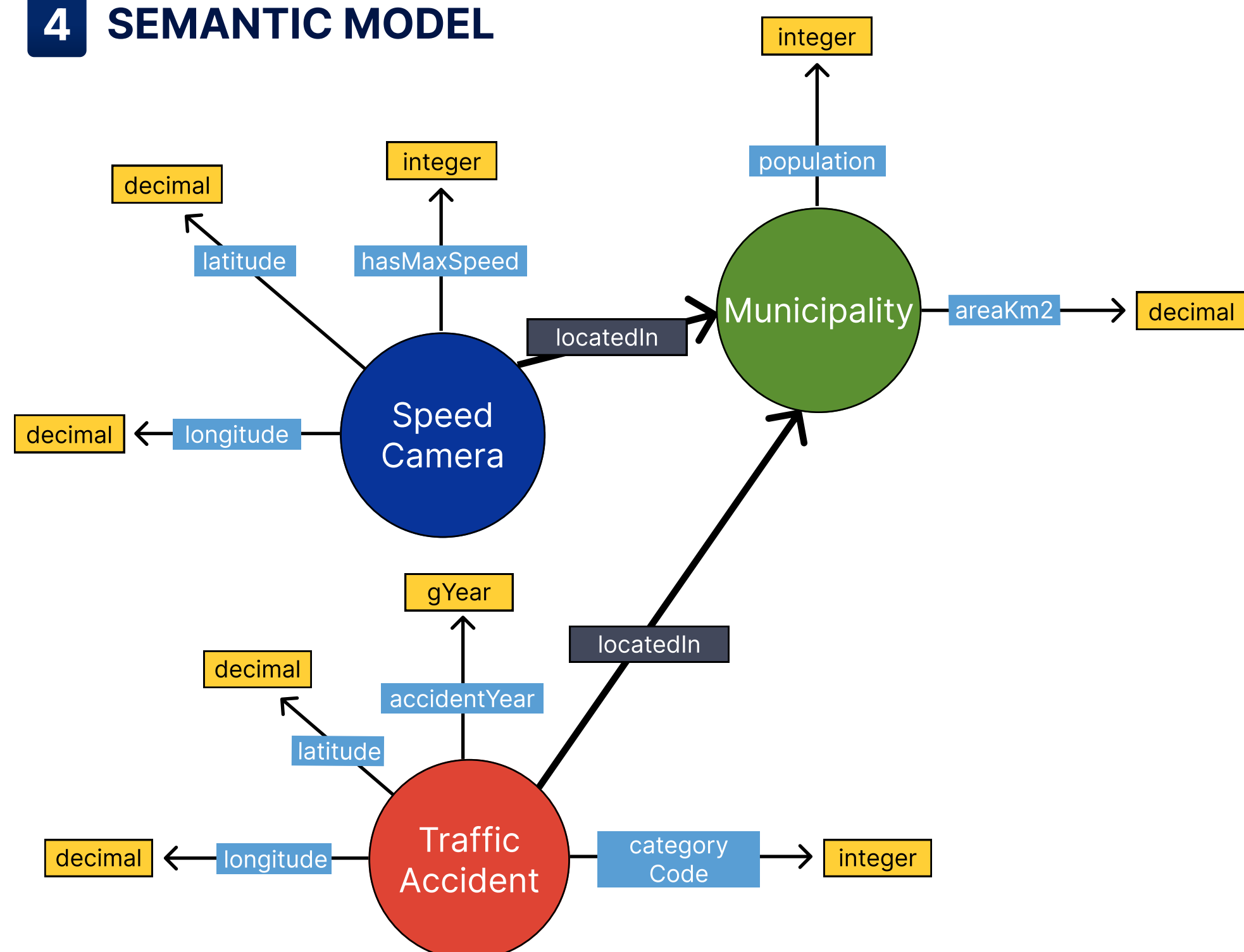
query and retrieve  
linked data



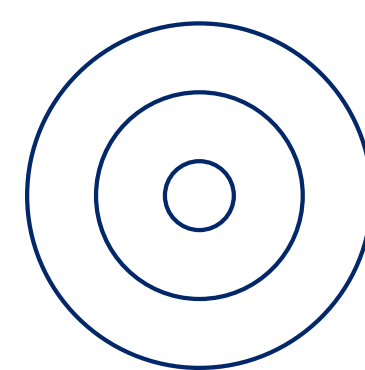
#### Visualization

maps, dashboards,  
exploration

### 4 SEMANTIC MODEL



### 5 RESEARCH GOAL



How can open data  
be linked to  
explore speed  
cameras and  
accidents in Saxony?

### 6 RDF OUTPUT EXAMPLES

#### 1 Source 1 (OpenStreet Map)

```
ex:speed-camera-node_11958280000
  a ex:SpeedCamera ;
  ex:direction "45" ;
  ex:hasMaxSpeed 50 ;
  ex:latitude 51.323478 ;
  ex:longitude 12.3649086 ;
  ex:osmId "node/11958280000" ;
  ex:source <https://www.openstreetmap.org/...> .
```

#### 2 Source 2 (Unfallatlas)

```
ex:accident-14240101119810955610
  a ex:TrafficAccident ;
  ex:accidentCategoryCode "3" ;
  ex:accidentMonth 1 ;
  ex:accidentYear "2024"^^xsd:gYear ;
  ex:involvesCar 1 ;
  ex:involvesMotorcycle 0 ;
  ex:latitude 50.68397282500007 ;
  ex:locatedIn ex:saxony ;
  ex:longitude 12.80218097200003 ; .
```

#### 3 Source 3 (Open Data Sachsen)

```
ex:saxony_area
  a ex:Dataset ;
  ex:name "Free State of Saxony" ;
  ex:areaSquareKilometers "18449.885380" .

ex:saxoy_vehicle_stock_2010
  a ex:Dataset ;
  ex:name "Freistaat Sachsen" ;
  ex:date "2010-01-01" ;
  ex:totalVehicles "2428237" ;
  ex:passengerCars "2061040" ;
  ex:trucks "162891" .
```

### 7 SAXONY OVERVIEW



**273**

Speed Cameras (fixed)  
from OpenSreetMap



**12,847**

Accident Records (2024)  
map layer in progress



**418**

Municipalities in Saxony  
population & area data

### 8 EXPECTED OUTCOME

A small RDF Knowledge Graph for traffic safety data in Saxony

The final project should connect speed cameras, accident records and regional context data in one graph. This enables simple SPARQL queries and visual exploration of patterns that would be harder to see in the original separate data sources.

#### GitHub Repository



Code, data extraction  
scripts and RDF output  
files